

Q	is any	Age	stu
♀		20	yes

$$P(\text{buy} = Y | \hat{x}) = P(\hat{x} | \text{buy} = Y) P(\text{buy} = \text{yes})$$

$$= \frac{2}{9} \times \frac{6}{9} \times \frac{9}{14}$$

$$= 0.095 \quad \#$$

$$P(\text{buy} = N | \hat{x}) = P(\hat{x} | \text{buy} = N) P(\text{buy} = N)$$

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14}$$

$$= 0.043 \quad \#$$

Yes $\#$